

Program9: Organize the Kubernetes Pods using the Labels and Namespaces

Step1: Create a file for namespaces.yaml

Step2: create a file as pods.yaml

Step3: apply the namespaces & pod yaml file

```
apiVersion: v1
kind: Namespace
Metadata:
  name: dev
—
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step4: verify the pods by namespaces

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
labels:
  app: frontend
  env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
labels:
  app: backend
  env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
---
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Step5: apply the namespaces and pods yaml file

```
Kubectl apply -f namespaces.yaml
Kubectl apply -f pods.yaml
```

Step6: verify pods by namespaces

```
Kubectl get pods -n dev
Kubectl get pods -n prod
```

Step7: check pod status

```
Kubectl get pods
```

Step8: filter pods by label

```
Kubectl get pods -n dev -l app=frontend
Kubectl get pods -n prod -l app=backend
```

```

1RV24MC036_gayatri_v_k@ubuntu:~$ kubectl get nodes
NAME          STATUS    ROLES    AGE   VERSION
minikube      Ready     control-plane  10m   v1.34.0
1RV24MC036_gayatri_v_k@ubuntu:~$ mkdir program9
1RV24MC036_gayatri_v_k@ubuntu:~$ cd program9
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ nano namespace.yaml
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ nano pods.yaml
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ minikube image load nginx
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ kubectl apply -f namespaces.yaml
error: the path "namespaces.yaml" does not exist
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ kubectl apply -f namespace.yaml
namespace/dev created
namespace/prod created
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           15s
frontend-pod   1/1     Running   0           15s
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           22s
frontend-pod   1/1     Running   0           22s
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0           31s
1RV24MC036_gayatri_v_k@ubuntu:~/program9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           39s

```